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Inventor(s)	Baldi; Pradeep et al.

Scenario planning solutions

Abstract

A system and method are disclosed for performing rough cut capacity planning. The method includes receiving supply chain transaction data as transaction tables, generating a base plan and generating updated transaction tables, denormalizing the base plan and the updated transaction tables, generating supply chain network flow paths and supply chain network data, solving a rough cut capacity planning problem based at least in part on the supply chain network flow paths, the supply chain network data and simulation data, repeating at least the generating and solving until business goals of the rough cut capacity planning meet a threshold, and updating the simulation data based on an upsert process. The method further includes relaxing supply chain network constraints, inverting the supply chain network, and traversing a perturbation in the supply chain network constraints as demands in a reverse direction towards customer nodes.

Inventors: Baldi; Pradeep (Irving, TX), Linder; Mathieu (Dallas, TX), Beringer; Joerg (Redwood City, CA), Vonorych; Anton (Ternopil, UA)

Applicant: Blue Yonder Group, Inc. (Scottsdale, AZ)

Family ID: 1000007016438

Assignee: Blue Yonder Group, Inc. (Scottsdale, AZ)

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Primary Examiner: Ade; Garcia

Attorney, Agent or Firm: Spencer Fane LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is related to that disclosed in the U.S. Provisional Application No. 63/344,892, filed May 23, 2022, entitled "Scenario Planning Solutions." U.S. Provisional Application No. 63/344,892 is assigned to the assignee of the present application. The present invention hereby claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/344,892.

TECHNICAL FIELD

(1) The present disclosure relates generally to supply chain planning, and more in particular relates to scenario planning solutions.

BACKGROUND

(2) Existing scenario planning solutions lack the ability to quickly generate multiple risk and/or reward perspectives to support supply chain decision making. These scenario planning solutions also lack practical user interface workflows that are needed to evaluate business trade-offs associated with different scenarios. In addition, these scenario planning solutions lack the ability to generate multiple point of views in order to support timely decision making. Furthermore, existing scenario planning solutions involve numerous manual inputs across multiple screens which makes their use a time consuming process prone to errors. Other shortcomings of existing scenario planning solutions is that they require tacit knowledge by someone who is configuring or building a scenario. That is, a user that is unable to set the right thresholds and the right parameters, can never achieve the desired output. Furthermore, it is not possible to orchestrate a scenario that spans across multiple solution areas and business processes, using existing scenario planning solutions. Lastly, if using cloud computing infrastructure to execute existing scenario planning solutions, the cost of evaluating scenarios may prohibitive from a business standpoint. Thus, existing scenario planning solutions are inflexible, time consuming, prone to errors, require expert knowledge to operate and may be prohibitively expensive to use, all of which is undesirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of the present invention may be derived by referring to the detailed description when considered in connection with the following illustrative figures. In the figures, like reference numbers refer to like elements or acts throughout the figures.
- (2) FIG. 1 illustrates a supply chain network, according to an embodiment;
- (3) FIG. 2 illustrates the supply chain planner of FIG. 1 in greater detail, according to an embodiment;
- (4) FIG. 3 illustrates a rough cut capacity planning (RCCP) method, according to an embodiment;
- (5) FIG. 4 illustrates a RCCP workflow, according to an embodiment;
- (6) FIG. 5 illustrates a RCCP preparation, according to an embodiment;
- (7) FIG. 6 illustrates a RCCP core logic, according to an embodiment;
- (8) FIG. 7 illustrates performance data of the RCCP method of FIG. 3, according to an embodiment;
- (9) FIG. 8 illustrates a performance chart, according to an embodiment;
- (10) FIG. 9 illustrates an example chart of RCCP data, according to an embodiment;
- (11) FIG. 10 illustrates a map-reduce type approach for the RCCP method of FIG. 3, according to an embodiment;
- (12) FIG. 11 illustrates a first user interface, according to an embodiment;
- (13) FIG. 12 illustrates a second user interface, according to an embodiment; and
- (14) FIG. 13 illustrates an example comparison chart, according to an embodiment.

DETAILED DESCRIPTION

(15) Aspects and applications of the invention presented herein are described below in the drawings and detailed description of the invention. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts.

(16) In the following description, and for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various aspects of the invention. It will be understood, however, by those skilled in the relevant arts, that the present invention may be practiced without these specific details. In other instances, known structures and devices are shown or discussed more generally in order to avoid obscuring the invention. In many cases, a description of the operation is sufficient to enable one to implement the various forms of the invention, particularly when the operation is to be implemented in software. It should be noted that there are many different and alternative configurations, devices and technologies to which the disclosed inventions may be applied. The full scope of the inventions is not limited to the examples that are described below.

(17) As described in more detail below, embodiments of the following disclosure provide a low-touch decision support framework to provide the best input variable values from among all possibilities without explicitly evaluating each possibility using simulation optimization for a fixed time or iterations using a combination of stochastic and heuristics techniques. According to embodiments, a simulation iteration is a test or a series of tests in which meaningful changes are made to the input variables so that we may observe and identify the reasons for changes to the output variable(s) or goals and then iteratively refine the set of input variables to get us closer to the goals.

(18) Embodiments, provide for a test of multiple input variables in order to observe and identify the change that their effect has on output variables or business goals. In addition, embodiments then iterate through them until they get closer to the business goals. That is, instead of asking users to provide definite inputs, embodiments look to the intent. Use of embodiments may reduce the

computational costs of executing supply chain scenario panning. In addition, or as an alternative, use of embodiments ensures that a simulation experiment does not become computationally prohibitive from a cost standpoint.

(19) According to embodiments and for supply chain planning applications, there are two broad categories of scenarios: 1) perturbation demand and the ability to see its impact on the network and 2) perturbation network parameters and the ability to see its impact on the demand. Typical supply chain solving techniques involve traversing the end customer demand up the network through various fulfillment stages and related constraints to meet a set of business objectives. This technique is better suited for demand perturbation-based scenarios and requires checking multiple levels of constraints across the supply chain network which could get very complex and computationally intensive. Through simulation-optimization, embodiments combine fast and iterative simulations with optimization goals to evaluate thousands of perturbations within a shorter time period than existing scenario planning solutions. Embodiments provide the ability to invert the network and identify a constraint that may be used as input while demand becomes the output. That is, embodiments traverse the network in a reverse way, that allows for the ability to respect the constraints without explicitly accounting for them. Embodiments provide for this to be achieved by relaxing the network constraints, inverting the network, and traversing the perturbation in network constraints as demands in the reverse direction toward the customer nodes.

(20) FIG. 1 illustrates supply chain network **100**, in accordance with a first embodiment. Supply chain network **100** comprises supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, one or more computers **160**, network **170**, and one or more communication links **180-190**.

Although a single supply chain planner **110**, a single inventory system **120**, a single transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, one or more computers **160**, and a single network **170** are shown and described, embodiments contemplate any number of supply chain planners, inventory systems, transportation management systems, imaging devices, supply chain entities, computers, or networks, according to particular needs.

(21) In one embodiment, supply chain planner **110** comprises server **112** and database **114**. Server **112** comprises one or more modules that model, generate, and solve a supply chain planning problem. As described in more detail below, supply chain planner **110** may model a supply chain planning problem and calculate a solution using one or more solvers, such as, for example, a Deep Tree solver, a MAP solver, an LP optimization solver, and the like. By way of example only and not by way of limitation, supply chain planner **110** may model a supply chain planning problem as a multi-objective hierarchical linear programming (LP) problem comprising an LP constraint-variable matrix and calculate the globally-optimal LP solution to the supply chain planning problem. Other embodiments of supply chain planner model a supply chain planning problem for solving by a Deep Tree solver, MAP solver, and other like planning problem solvers.

(22) Inventory system **120** comprises server **122** and database **124**. Server **122** of inventory system **120** is configured to receive and transmit product data **216** (see FIG. 2) (including, for example, item identifiers, pricing data, and attribute data), inventory data **220** (see FIG. 2) (including, for example, inventory levels), and other like data about one or more items at one or more locations in supply chain network **100**. Server **122** stores and retrieves data about the one or more items from database **124** or from one or more locations in supply chain network **100**.

(23) Transportation management system **130** comprises server **132** and database **134**. According to embodiments, transportation management system **130** directs the one or more transportation vehicles **136** to ship one or more items between one or more supply chain entities **150**, based, at least in part, on the number of items currently in transit in transportation management system **130**, a supply chain plan, such as, for example, a supply chain master plan, the number of items currently in stock at one or more supply chain entities **150**, a forecasted demand, a supply chain

disruption, a material or capacity reallocation, current and projected inventory levels at one or more stocking locations, and/or one or more additional factors described herein. One or more transportation vehicles **136** comprise, for example, any number of trucks, cars, vans, boats, airplanes, unmanned aerial vehicles (UAVs), cranes, robotic machinery, or the like. One or more transportation vehicles **136** may comprise radio, satellite, or other communication that communicates location information (such as, for example, geographic coordinates, distance from a location, global positioning satellite (GPS) information, or the like) with supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and/or one or more supply chain entities **150** to identify the location of one or more transportation vehicles **136** and the location of an item of any inventory or shipment located on one or more transportation vehicles **136**.

(24) One or more imaging devices **140** comprise one or more processors **142**, memory **144**, one or more sensors **146**, and may include any suitable input device, output device, fixed or removable computer-readable storage media, or the like. According to embodiments, one or more imaging devices **140** comprise an electronic device that receives data from one or more sensors **146**. One or more sensors **146** of one or more imaging devices **140** may comprise an imaging sensor, such as, a camera, scanner, electronic eye, photodiode, charged coupled device (CCD), or any other electronic component that detects visual characteristics (such as color, shape, size, fill level, or the like) of objects. One or more imaging devices **140** may comprise, for example, a mobile handheld electronic device such as, for example, a smartphone, a tablet computer, a wireless communication device, and/or one or more networked electronic devices configured to image items using one or more sensors **146** and transmit product images to one or more databases.

(25) In addition, or as an alternative, one or more sensors **146** may comprise a radio receiver and/or transmitter configured to read from and/or write to an electronic tag, such as, for example, a radio-frequency identification (RFID) tag. Each item may be represented in supply chain network **100** by an identifier, including, for example, Stock-Keeping Unit (SKU), Universal Product Code (UPC), serial number, barcode, tag, RFID, or like device that encodes identifying information. One or more imaging devices **140** may generate a mapping of one or more items in supply chain network **100** by scanning an identifier or device associated with an item and identifying the item based, at least in part, on the scan. This may include, for example, a stationary scanner located at one or more supply chain entities **150** that scans items as the items pass near the scanner. As explained in more detail below, supply chain planner **110**, inventory system **120**, transportation management system **130**, and one or more imaging devices **140** may use the mapping of an item to locate the item in supply chain network **100**. The location of the item may be used to coordinate the storage and transportation of items in supply chain network **100** according to one or more plans generated by supply chain planner **110** and/or a reallocation of materials or capacity. Plans may comprise one or more of a master supply chain plan, production plan, distribution plan, and the like.

(26) Additionally, one or more sensors **146** of one or more imaging devices **140** may be located at one or more locations local to, or remote from, one or more imaging devices **140**, including, for example, one or more sensors **146** integrated into one or more imaging devices **140** or one or more sensors **146** remotely located from, but communicatively coupled with, one or more imaging devices **140**. According to some embodiments, one or more sensors **146** may be configured to communicate directly or indirectly with one or more of supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, one or more computers **160**, and/or network **170** using the one or more communication links **180-190**.

(27) As shown in FIG. 1, supply chain network **100** comprising supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and one or more supply chain entities **150** may operate on one or more computers **160** that are integral to or separate from the hardware and/or software that support supply chain planner **110**,

inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and one or more supply chain entities **150**. One or more computers **160** may include any suitable input device **162**, such as a keypad, mouse, touch screen, microphone, or other device to input information. Output device **164** may convey information associated with the operation of supply chain network **100**, including digital or analog data, visual information, or audio information.

(28) One or more computers **160** may include fixed or removable computer-readable storage media, including a non-transitory computer-readable medium, magnetic computer disks, flash drives, CD-ROM, in-memory device or other suitable media to receive output from and provide input to supply chain network **100**. One or more computers **160** may include one or more processors **166** and associated memory to execute instructions and manipulate information according to the operation of supply chain network **100** and any of the methods described herein. In addition, or as an alternative, embodiments contemplate executing the instructions on one or more computers **160** that cause one or more computers **160** to perform functions of the method. An apparatus implementing special purpose logic circuitry, for example, one or more field programmable gate arrays (FPGA) or application-specific integrated circuits (ASIC), may perform functions of the methods described herein. Further examples may also include articles of manufacture including tangible non-transitory computer-readable media that have computer-readable instructions encoded thereon, and the instructions may comprise instructions to perform functions of the methods described herein.

(29) Supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and one or more supply chain entities **150** may each operate on one or more separate computers **160**, a network of one or more separate or collective computers **160**, or may operate on one or more shared computers **160**. In addition, supply chain network **100** may comprise a cloud-based computing system having processing and storage devices at one or more locations local to, or remote from, supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and one or more supply chain entities **150**. In addition, each of one or more computers **160** may be a work station, personal computer (PC), network computer, notebook computer, tablet, personal digital assistant (PDA), cell phone, telephone, smartphone, mobile device, wireless data port, augmented or virtual reality headset, or any other suitable computing device. In an embodiment, one or more users may be associated with supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and one or more supply chain entities **150**. These one or more users may include, for example, a “manager” or a “planner” handling supply chain planning and/or one or more related tasks within supply chain network **100**. In addition, or as an alternative, these one or more users within supply chain network **100** may include, for example, one or more computers **160** programmed to autonomously handle, among other things, production planning, demand planning, option planning, sales and operations planning, supply chain master planning, plan adjustment after supply chain disruptions, order placement, automated warehouse operations (including removing items from and placing items in inventory), robotic production machinery (including producing items), and/or one or more related tasks within supply chain network **100**.

(30) One or more supply chain entities **150** may represent one or more suppliers **152**, manufacturers **154**, distribution centers **156**, and retailers **158** of supply chain network **100** or one or more supply chain networks **100**, including one or more enterprises. One or more suppliers **152** may be any suitable entity that offers to sell or otherwise provides one or more components to one or more manufacturers **154**. One or more suppliers **152** may, for example, receive a product from a first supply chain entity in supply chain network **100** and provide the product to another supply chain entity. One or more suppliers **152** may comprise automated distribution systems **153** that automatically transport products to one or more manufacturers **154** based, at least in part, on the number of items currently in transit in transportation management system **130**, a supply chain plan, the number of items currently in stock at one or more supply chain entities **150**, a forecasted

demand, a supply chain disruption, a material or capacity reallocation, current and projected inventory levels at one or more stocking locations, and/or one or more additional factors described herein.

(31) One or more manufacturers **154** may be any suitable entity that manufactures at least one product. One or more manufacturers **154** may use one or more items during the manufacturing process to produce any manufactured, fabricated, assembled, or otherwise processed item, material, component, good or product. Items may comprise, for example, components, materials, products, parts, supplies, or other items, that may be used to produce products. In addition, or as an alternative, an item may comprise a supply or resource that is used to manufacture the item, but does not become a part of the item. In one embodiment, a product represents an item ready to be supplied to, for example, another supply chain entity, an item that needs further processing, or any other item. One or more manufacturers **154** may, for example, produce and sell a product to one or more suppliers **152**, another one or more manufacturers **154**, one or more distribution centers **156**, one or more retailers **158**, or any other suitable customer or one or more supply chain entities **150**. One or more manufacturers **154** may comprise automated robotic production machinery **155** that produce products based, at least in part, on the number of items currently in transit in transportation management system **130**, a supply chain plan, the number of items currently in stock at one or more supply chain entities **150**, a forecasted demand, a supply chain disruption, a material or capacity reallocation, current and projected inventory levels at one or more stocking locations, and/or one or more additional factors described herein.

(32) One or more distribution centers **156** may be any suitable entity that offers to sell or otherwise distributes at least one product to one or more retailers **158**, customers, or any suitable one or more supply chain entities **150**. One or more distribution centers **156** may, for example, receive a product from a first supply chain entity in supply chain network **100** and store and transport the product for a second supply chain entity. One or more distribution centers **156** may comprise automated warehousing systems **157** that automatically transport an item to, remove an item from, or place an item into inventory of one or more retailers **158**, customers, or one or more supply chain entities **150** based, at least in part, on the number of items currently in transit in transportation management system **130**, a supply chain plan, the number of items currently in stock at one or more supply chain entities **150**, a forecasted demand, a supply chain disruption, a material or capacity reallocation, current and projected inventory levels at one or more stocking locations, and/or one or more additional factors described herein.

(33) One or more retailers **158** may be any suitable entity that obtains one or more products to sell to one or more customers. In addition, one or more retailers **158** may sell, store, and supply one or more components and/or repair a product with one or more components. One or more retailers **158** may comprise any online or brick and mortar location, including locations with shelving systems **159**. Shelving systems **159** may comprise, for example, various racks, fixtures, brackets, notches, grooves, slots, or other attachment devices for fixing shelves in various configurations. These configurations may comprise shelving with adjustable lengths, heights, and other arrangements, which may be adjusted by an employee of one or more retailers based on computer-generated instructions or automatically by machinery to place products in a desired location, and which may be based, at least in part, on the number of items currently in transit in transportation management system **130**, a supply chain plan, the number of items currently in stock at one or more supply chain entities **150**, a forecasted demand, a supply chain disruption, a material or capacity reallocation, current and projected inventory levels at one or more stocking locations, and/or one or more additional factors described herein.

(34) Although one or more suppliers **152**, manufacturers **154**, distribution centers **156**, and retailers **158** are shown and described as separate and distinct entities, the same entity may simultaneously act as any the one or more suppliers **152**, manufacturers **154**, distribution centers **156**, and retailers **158**. For example, one or more manufacturers **154** acting as a manufacturer could produce a

product, and the same entity could act as one or more suppliers **158** to supply a product to another one or more supply chain entities **150**. Although one example of supply chain network **100** is shown and described, embodiments contemplate any configuration of supply chain network **100**, without departing from the scope of the present disclosure.

(35) In one embodiment, supply chain planner **110** may be coupled with network **170** using communication link **180**, which may be any wireline, wireless, or other link suitable to support data communications between supply chain planner **110** and network **170** during operation of supply chain network **100**. Inventory system **120** may be coupled with network **170** using communication link **182**, which may be any wireline, wireless, or other link suitable to support data communications between inventory system **120** and network **170** during operation of supply chain network **100**. Transportation management system **130** may be coupled with network **170** using communication link **184**, which may be any wireline, wireless, or other link suitable to support data communications between transportation management system **130** and network **170** during operation of supply chain network **100**. One or more imaging devices **140** are coupled with network **170** using communication link **186**, which may be any wireline, wireless, or other link suitable to support data communications between one or more imaging devices **140** and network **170** during operation of supply chain network **100**. One or more supply chain entities **150** may be coupled with network **170** using communication link **188**, which may be any wireline, wireless, or other link suitable to support data communications between one or more supply chain entities **150** and network **170** during operation of supply chain network **100**. One or more computers **160** may be coupled with network **170** using communication link **190**, which may be any wireline, wireless, or other link suitable to support data communications between one or more computers **160** and network **170** during operation of supply chain network **100**. Although one or more communication links **180-190** are shown as generally coupling supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160** to network **170**, each of supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160** may communicate directly with each other, according to particular needs.

(36) In another embodiment, network **170** includes the Internet and any appropriate local area networks (LANs), metropolitan area networks (MANs), or wide area networks (WANs) coupling supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160**. For example, data may be maintained local to, or external of, supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160** and made available to one or more associated users of supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160** using network **170** or in any other appropriate manner. For example, data may be maintained in a cloud database at one or more locations external to supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160** and made available to one or more associated users of supply chain planner **110**, inventory system **120**, transportation management system **130**, one or more imaging devices **140**, one or more supply chain entities **150**, and one or more computers **160** using the cloud or in any other appropriate manner. Those skilled in the art will recognize that the complete structure and operation of network **170** and other components within supply chain network **100** are not depicted or described. Embodiments may be employed in conjunction with known communications networks and other components.

(37) FIG. 2 illustrates supply chain planner **110** of FIG. 1 in greater detail, in accordance with an embodiment. As discussed above, supply chain planner **110** comprises server **112** and database **114**.

Although supply chain planner **110** is shown as comprising a single server **112** and a single database **114**, embodiments contemplate any suitable number of servers or databases internal to, or externally coupled with, supply chain planner **110**.

(38) Server **112** of supply chain planner **110** may comprise modeler **202**, solver **204**, denormalization module **206**, simulation listener **208**, event hub **210**, RCCP solver **212**, upsert module **214**, and user interface module **216**. Although server **112** is shown and described as comprising a single modeler **202** a single solver **204**, a single denormalization module **206**, a single prep module **208**, a single event hub **210**, a single RCCP solver **212**, a single upsert module **214**, and a single user interface module **216**, embodiments contemplate any number of combination of modelers, solvers, denormalization modules, simulation listeners, event hubs, RCCP solvers, upsert modules, and user interface modules at one or more locations, local to, or remote from supply chain planner **110**, at one or more servers **112** or one or more computers **160** at any location in supply chain network **100**.

(39) Modeler **202** may model one or more supply chain planning problems of supply chain network **100**. According to embodiments, modeler **202** identifies resources, operations, buffers, and pathways, and maps supply chain network **100** as data models. For example, when using an LP solver, modeler **202** models a single- or multi-period supply chain planning problem that represents supply chain network **100** as a hierarchical multi-objective LP supply chain master planning problem comprising mathematical objective functions that represent business objectives, mathematical constraints that represent supply chain constraints, and lower and/or upper bounds on decision variables representing the supply chain data.

(40) Solver **204** comprises generates a solution to the supply chain planning problem using data from one or more of the supply chain data models. Solver **204** of server **112** comprises one or more optimization, heuristic, or mathematical solvers that utilize the data models to generate a solution to the supply chain planning problem. The solutions produced by solver **204** may be called a supply chain plan or base plan **220**, according to embodiments.

(41) Denormalization module **206** performs denormalization on transaction data **218**, such as transaction tables associated with a base supply chain plan as generated by solver **204**. For example, denormalization may be performed by adding redundant data to transaction data **218** in order to improve the read performance of transaction data **218**. In embodiments, transaction data **218** may have been previously normalized by a module of supply chain planner **110** configured to normalize data associated with supply chain plans. In embodiments, denormalizing transaction data **218** may be considered preparation for a rough cut capacity planning (RCCP) process.

(42) Prep module **208** performs various preparatory tasks in order to allow RCCP solver **212** to solve an RCCP problem. For example, may generate dataframe **222** from denormalized transaction data **218**. Prep module **208** may then use dataframe **222** to determine flow paths **224** and network data **226**. As described in further detail below, network data **226** may comprise a network of nodes representing supply chain network **100**, and flow paths **224** may include single paths from consumer to upstream nodes of network data **226**. Prep module **208** also includes a simulation listener, which continuously listens for simulation data **230** received from event hub **210**. Prep module **208** may then combine flow paths **224** with simulation data **230** to generate simulations for RCCP solver **212** to solve.

(43) Event hub **210** oversees the actions of prep module **208**, RCCP solver **212** and upsert module **214**, in order to correlate their outputs and feedback their output to iteratively improve the simulation process. For example, event hub **210** may receive flow paths **224** including network data **225** and simulation data **230** from prep module **208**, RCCP data **228** from RCCP solver **212** and upsert data from upsert module **214**. Event hub **210** may then combine the various data streams from prep module **208**, RCCP solver **212** and upsert module **214** to generate simulation data **230**. Simulation data **230** may then be provided to a simulation listener associated with prep module **208**, in order for simulation data **230** to be used in performing simulations solved by RCCP solver

212.

(44) RCCP solver **212** uses various serverless RCCP functions to perform an RCCP analysis using flow paths **224** including network data **225** and simulation data **230**. The result of the RCCP analysis is RCCP data **228**. In general, the RCCP analysis may be performed to determine an estimate of production that should be generated in order to meet simulated demand. Flow paths **224**, when provided to RCCP solver **212**, may include simulation data **230**. In such cases, solving flow paths **224** to generate RCCP data **228** may be considered solving a simulation based on constraints of the simulation, which is traversed bottom to top to determine a scenario most likely to meet the business objectives of supply chain network **110**.

(45) Upsert module **214** performs upserting of database **114** based on RCCP data **228**. For example, upsert module **214** may update a corresponding entry in database **114** if such an entry exists, or may insert a new entry in database **114** if no corresponding entry exists. In embodiments, upsert module **214** also provides RCCP data **228** to event hub **210** so that simulation data **230** may be generated.

(46) User interface module **216** provides interactive graphical elements comprising selectable elements that, in response to a user selection, initiate a predetermined action, such as, for example, selecting a simulation generated by RCCP solver **212** and initiate incremental planning of one or more scenarios, persisting planned scenario to database **114**, and other like actions, as described herein. In addition, user interface module **216** generates a user interface for selecting, visualizing, modifying, saving, and/or deleting one or more of: transaction data **218**, base plan **220**, dataframe **222**, flow paths **224**, network data **225**, RCCP data **228**, simulation data **230**, demand data **232**, inventory data **234**, supply chain models **236**, and inventory policies **238**, and the like. The user interface may display one or more visual elements on an associated display device including for example, data, data models, product images, attributes and attribute values, selectable time periods, and the like. In addition, the user interface displays data and interactive visual elements for selecting and configuring scenarios and plans of supply chain planner **110**.

(47) Database **114** of supply chain planner **110** may comprise one or more databases **114** or other data storage arrangement at one or more locations, local to, or remote from, server **112**. Database **114** comprises, for example, transaction data **218**, base plan **220**, dataframe **222**, flow paths **224**, network data **225**, RCCP data **228**, simulation data **230**, demand data **232**, inventory data **234**, supply chain models **236**, and inventory policies **238**. Although database **114** is shown and described as comprising transaction data **218**, base plan **220**, dataframe **222**, flow paths **224**, network data **225**, RCCP data **228**, simulation data **230**, demand data **232**, inventory data **234**, supply chain models **236**, and inventory policies **238**, embodiments contemplate any suitable number or combination of these at one or more locations, local to, or remote from, supply chain planner **110** according to particular needs.

(48) Transaction data **218** comprises data associated with supply chain network **100** corresponding to transactions of supply chain network **100**. Transaction data **218** may include sourcing data, network data, BOM data, ProductionMethod data, CustOrder data, or FcstOrder data.

(49) Base plan **220** comprises a base supply chain plan generated by solver **204**. For example, base plan **220** may comprise various plans for raw material distribution, manufacturing, storage, transportation or any other supply chain activity comprising actions from pre-production to retailing. Base plan **220** may be batched to generated transaction tables, which may be stored as transaction data **218**.

(50) Dataframe **222** comprises data which has been denormalized by denormalization module **206**. Dataframe **222** may include a set of nodes of a supply chain model, as well as a set of edges of a supply chain model, stored as separate objects. Dataframe **222** may be used to generate flow paths **224**, by combining the set of nodes and the set of edges into a complete network model.

(51) Flow paths **224** comprise paths through network graphs of supply chain network **100** generated by prep module **208**. In embodiments, flow paths **224** may be used by RCCP solver **212**

to generate scenarios for supply chain network **100**, as described in further detail below.

(52) Network data **226** comprises data of a supply chain model. Network data **226** may comprise various nodes and edges associated with the supply chain model. For example, and as described in further detail below, a node of network data **226** may correspond to a particular distribution center or a particular retailer of supply chain network **100**, while an edge of network data **226** may correspond to the connection between the distribution center and the retailer.

(53) RCCP data **228** comprises solved RCCP data, such as supply chain scenarios. For example, when performing an RCCP analysis, RCCP data **228** may comprise a set of scenarios analyzed by RCCP solver **212** ranked by a particular metric, such as total revenue or any other relevant metric. That is, RCCP data **228** may comprise the result of many hundreds or thousands of simulations indicating which supply chain scenarios are most likely to result in completion of business objectives of supply chain network **100**.

(54) Simulation data **230** comprises various levers or adjustments made to supply chain data to determine the likely results from adjusting the levers. For example, an increase to production at a particular factory may be a lever used to generate a simulation, by simulating the likely impact on supply within the supply chain network. Other levers of simulation data **230** may include adjustments to raw material availability, product demand, transportation costs or any other metric bearing on the output of supply chain network **100**.

(55) Demand data **218** may comprise, for example, any data relating to past sales, past demand, purchase data, promotions, events, or the like of one or more supply chain entities **150**. Demand data **218** may cover a time interval such as, for example, by the minute, hour, daily, weekly, monthly, quarterly, yearly, or any suitable time interval, including substantially in real time. According to embodiments, demand data **218** may include historical demand and sales data or projected demand forecasts for one or more retail locations, customers, regions, or the like of one or more supply chain entities **150** and may include historical or forecast demand and sales segmented according to product attributes, customers, regions, or the like.

(56) Inventory data **220** may comprise any data relating to current or projected inventory quantities or states, order rules, or the like. For example, inventory data **220** may comprise the current level of inventory for each item at one or more stocking locations across supply chain network **100**. In addition, inventory data **220** may comprise order rules that describe one or more rules or limits on setting an inventory policy, including, but not limited to, a minimum order quantity, a maximum order quantity, a discount, a step-size order quantity, and batch quantity rules. According to some embodiments, supply chain planner **110** accesses and stores inventory data **220** in database **114**, which may be used by supply chain planner **110** to place orders, set inventory levels at one or more stocking points, initiate manufacturing of one or more items (or components of one or more items), or the like. In addition, or as an alternative, inventory data **220** may be updated by receiving current item quantities, mappings, or locations from inventory system **120**, transportation management system **130**, one or more imaging devices **140**, and/or one or more supply chain entities **150**.

(57) Supply chain models **222** may comprise characteristics of a supply chain setup to deliver the customer expectations of a particular customer business model. These characteristics may comprise differentiating factors, such as, for example, MTO (Make-to-Order), ETO (Engineer-to-Order) or MTS (Make-to-Stock). Additionally, or in the alternative, supply chain models **222** may comprise characteristics that specify the supply chain structure in even more detail, including, for example, specifying the type of collaboration with the customer (e.g. Vendor-Managed Inventory (VMI)), from which stocking locations or suppliers items may be sourced, customer priorities, demand priorities, how products may be allocated, shipped, or paid for, by particular customers, and the destination stocking locations or one or more supply chain entities **150** where items may be transported. Each of these characteristics may lead to different supply chain models **222**.

(58) Inventory policies **224** may comprise any suitable inventory policy describing the reorder point and target quantity, or other inventory policy parameters that set rules for supply chain

planner **110** to manage and reorder inventory. Inventory policies **224** may be based on target service level, demand, cost, fill rate, or the like. According to embodiment, inventory policies **224** comprise target service levels that ensure that a service level of one or more supply chain entities **150** is met with a certain probability. For example, one or more supply chain entities **150** may set a target service level at 95%, meaning one or more supply chain entities **150** will set the desired inventory stock level at a level that meets demand 95% of the time. Although, a particular target service level and percentage is described; embodiments contemplate any target service level, for example, a target service level of approximately 99% through 90%, 75%, or any target service level, according to particular needs. Other types of service levels associated with inventory quantity or order quantity may comprise, but are not limited to, a maximum expected backlog and a fulfillment level. Once the service level is set, supply chain planner **110** may determine a replenishment order according to one or more replenishment rules, which, among other things, indicates to one or more supply chain entities **150** to determine or receive inventory to replace the depleted inventory.

(59) FIG. 3 illustrates RCCP method **300**, according to an embodiment. RCCP method **300** comprises one or more activities, which although described in a particular order, may be performed in one or more permutations according to particular needs.

(60) At activity **302**, supply chain planner **110** receives transaction data **218**. According to embodiments, supply chain planner **110** receives transaction data **218** from database **114**, one or more supply chain entities **150**, and/or one or more locations local to, or remote from, supply chain network **100**. By way of example only and not by way of limitation, RCCP method **300** is described in connection with example workflow **400** of FIG. 4. At activity **302**, transaction data **218** is received as transaction tables **410** by modeler **202** in base plan batch process **402**.

(61) At activity **304**, supply chain planner **110** generates base plan **220**. As disclosed above, modeler **202** generates a supply chain model representing a supply chain problem, and solver **204** solves the supply chain problem to generate a supply chain plan comprising base plan **220**. Continuing with the previous example, base plan **220** is generated during base plan batch process **402** and generates updated transaction tables **412**. By way of example and not by way of limitation, base plan batch process **402** prepares transaction tables **412** for a demand perturbation scenario or a network perturbation scenario for RCCP preparation **404**.

(62) At activity **306**, denormalization module **206** performs denormalization on base plan **220** and transaction data **218**. Continuing with the example of RCCP workflow **400**, denormalization module **206** receives updated transaction tables **412** to generate dataframe **222** in RCCP preparation **404**. According to an embodiment, denormalization module **204** may perform a denormalization process such as database management system (DBMS) support, database administrator (DBA) implementation, or other suitable denormalization process.

(63) At activity **308**, modeler **202** generates flow paths **224** and network data **226** based at least in part on dataframe **222**. In the example of RCCP workflow **400**, prep module **208** generates flow paths **224** with network **226** from dataframe **222**, in RCCP preparation **404**.

(64) At activity **310**, prep module **208** monitors flow paths **224**, network data **226**, and simulation data **230**. As described in greater detail below, prep module **208** monitors data during RCCP method **300**, providing for iterative simulations. Continuing with the previous example, prep module **208** receives flow paths **224** and network data **226** while monitoring for simulation data **230**, in RCCP preparation **404**.

(65) At activity **312**, prep module **208** provides flow paths **224**, network data **226**, and simulation data **230** to RCCP solver **212**. In the RCCP workflow **400** example, prep module **208** provides flow paths **224**, network data **226**, and simulation data **230**, to RCCP solver **212** of RCCP serverless compute **406**. When simulation data **230** is not available, such as on the first iteration of an iterative simulation, prep module **208** may provide flow paths **224** solely based on network data **226** as an alternative. In addition, or as an alternative, prep module **208** provides flow paths **224**, network

data **226**, and simulation data **230** from RCCP serverless compute **406** to database **114** and/or event hub **210**.

(66) At activity **314**, RCCP solver **212** solves the data received from simulation listener **208**. Continuing with the RCCP workflow **400** example, RCCP solver **212** solves the data received from simulation listener **208**, in RCCP serverless compute **406**. According to an embodiment, RCCP solver **212** traverses supply chain network **100** in a reverse manner. Although RCCP solver **210** is shown and described as residing in server **112**, embodiments contemplate RCCP solver **210** residing solely in memory associated with one or more computers **160** at one or more locations local to or remote from supply chain network **100**.

(67) At activity **316**, RCCP solver **212** generates RCCP data **228**. In the RCCP workflow **400** example, RCCP solver **212** generates RCCP data **228**, in RCCP serverless compute **406**.

(68) At activity **318**, RCCP solver **212** transmits RCCP data **228** to upserter module **214**. Continuing with the previous example, RCCP solver **212** transmits RCCP data **228** to upserter module **214** of upserter service **408**. In addition, or as an alternative, RCCP solver **212** transmits RCCP data **228** to event hub **210**.

(69) At activity **320**, upserter module **214** executes upsert process on RCCP data **228**. In the example of RCCP workflow **400**, upserter module **214** executes upsert process in upserter service **408**. Continuing this example, upserter service **408** executes and transmits RCCP simulation finish event to event hub **210**, as shown in FIG. **10**. In addition, or as an alternative, upserter service **408** updates database **114**.

(70) At activity **322**, upsert module **214** updates simulation data **230**. Continuing with the RCCP workflow **400** example, upsert module **214** updates simulation data **230**, which is received by simulation listener **208**, as described above with activity **310**.

(71) FIG. **4** illustrates RCCP workflow **400**, according to an embodiment. RCCP workflow **400** comprises base plan batch process **402**, RCCP preparation **404**, RCCP serverless compute **406**, and upserter service **408**. As shown, RCCP workflow **400** may be initiated by computer **160** executing a GUI. Computer **160** may initiate a simulation within event hub **210**. In embodiments, computer **160** may be used to directly update the data of database **114**.

(72) At base plan batch process **402**, transaction tables **410** are processed by abase plan function to generate updated transaction tables **412**. Transaction tables **412** may be provided to RCCP preparation **404**, as performed by prep module **208**. Within RCCP preparation **404**, denormalization module **206** performs denormalization to generate dataframe **222**. Then, prep module **208** uses dataframe **222** to generate flow paths **224**. Flow paths **224** include network data **226** comprising data of a graph model used to model supply chain network **110**. RCCP preparation **404** may also include a simulation listener configured to continuously listen for simulation data **230** received from event hub **210**. In embodiments, such as a first run of an iterative simulation process, simulation listener may provide flow paths **224** to event hub **210** and RCCP serverless compute **406**. In other embodiments, the simulation listener may combine flow paths **224** with simulation data **230**, and pass the combined flow paths **224** to event hub **210** and RCCP serverless compute **406**. During RCCP preparation **404**, prep module **208** may also update database **114** with flowpaths **224** or simulation data **230**.

(73) At RCCP serverless compute **206**, one or more RCCP solvers **212** use flow paths **224**, with or without simulation data **230** as described above, to generate one or more scenario results as RCCP data **228**. RCCP data **228** may be provide to upserter service **408** as well as event hub **210**. Upserter service **408** may store RCCP data **228** in database **114** as appropriate and may also provide RCCP data **228** to event hub **210**. Event hub **210** correlates the various data from computer **160**, prep module **208** and upserter service **408** to generate simulation data **230**. For example, computer **160** may transmit specified changes to levers received as input to event hub **210**, and event hub **210** may thereafter provide to changes to the levers as simulation data **230**.

(74) FIG. **5** illustrates RCCP preparation **404**, according to an embodiment. RCCP preparation **404**

processes transaction tables **410** and dataframes **222** in sixteen seconds for example data of a paint manufacturer on a laptop computer with sixteen gigabytes of memory. Continuing with this example, RCCP preparation **404** processes dataframes **222** into network graph of supply chain **502**, and processes network graph of supply chain **502** into independent flow paths **224** in fifty two seconds.

(75) Transaction tables **410** may include various data tables, such as a sourcing table or a network table, as illustrated. Transaction tables **410** may be combined by prep module **208** to generate dataframe **222**, comprising a node dataframe and an edge dataframe. Prep module **208** may thereafter use dataframe **222** to generate a complete network graph of supply chain network **100**, illustrated as network graph of supply chain **502**. Independent flow paths **224** represent particular end-to-end traversals of network graph of supply chain **502**. For example, every possible path from first node to an ending node may be stored as independent flow paths **224**.

(76) FIG. **6** illustrates RCCP core logic **600**, according to an embodiment. RCCP core logic **600** traverses flow paths **222** at each node and each edge. Flow paths **222** comprise a network of nodes **602-614** connected by edges **616-626**. According to embodiments, RCCP solver **212** performs the ask_node process by checking for available material, and when not available, performing the ask_edge process on the upstream edges, waiting for return of ask_node process, and returning extra material needed. RCCP solver **212** performs the ask_edge process by checking for lead time, finding source node with lead time, performing the ask node process on source node, and returning the results from upstream. Embodiments contemplate that RCCP core logic **600** considers rule based sourcing, proportional sourcing, push based planning, and the like.

(77) As an example, node **602** corresponds to the inventory availability for a particular finished good (fg_2) at a customer, represented as fg_2_at_cust. In order to determine the demand for fg_2, RCCP solver **212** performs the ask_node process at fg_2_at_cust. Because the demand is not yet calculated, the ask_node process calls the ask_edge process for edge **616**. The ask_edge process for edge **616** calls the ask_node process of its source node, in this case node **604**, corresponding to the inventory of fg_2 at a particular distribution center, represented as fg_2_at_dc. The ask_node and ask_edge processes continue to call each other in this fashion until the end of the flow path is reached. In this example, the final node is either raw material availability at plant 2 (rm_at_plant_2 of node **608**) or the raw material availability at a supplier (rm_at_supplier of node **614**). In either case, RCCP solver **212** determines the raw material availability, and then calculates backwards back to node **602**, in order to determine the estimated inventory availability of fg_2 to the customer.

(78) FIG. **7** illustrates performance data **700** of RCCP method **300**, according to an embodiment. Performance data **700** comprises a chart showing the total time in milliseconds (Time_Taken) for various combinations of ask levels (Levels_To_Ask) and simulation quantities

(Num_Simulations). Result **702** indicates that even for sixteen ask levels and one thousand simulations, the time taken is approximately 2028 milliseconds. Embodiments contemplate running one thousand simulations in less than 0.5 seconds when running RCCP method **300** in parallel.

(79) FIG. **8** illustrates performance chart **800**, according to an embodiment. Chart **800** shows total time (y-axis **802**) for various quantities of simulations (x-axis **804**) and various quantities of ask levels (one level **806**, two level **808**, four level **810**, eight level **812**, and sixteen level **814**).

(80) FIG. **9** illustrates example chart **900** of RCCP data **228**, according to an embodiment. Chart **900** shows results of RCCP solving the plan using RCCP method **300**. According to embodiments, KPIs and BSC are derived from RCCP data **228**, such as, for example, total cost to serve, total manufacturing cost, total transportation cost, total procurement cost, total revenue, total profit, and the like.

(81) FIG. **10** illustrates a map-reduce type approach **1000** for RCCP method **300**, according to an embodiment. Map-reduce type approach **1000** comprises iterative steps allowing for the fulfillment of business objectives to be completed. Map-reduce type approach **1000** states with insights to

RCCP **1002** and various levers **1** through **m** **1004** being provided to an OBO ask generator **1006**. OBO ask generator **1006** generates various OBO asks **1008**. OBO asks **1008** can be provided to RCCP service **1010**, such as RCCP solver **212**, in order for OBO asks **1008** to be solved. The solutions produced by RCCP service **1010** are plans **1012**.

(82) Plans **1012** may be analyzed by goal inspector **1016**. Goal inspector **1016** may determine, if the goals defined in OBO asks **1008** have not been sufficiently achieved, such as by failing to meet a certain success threshold, to refine one or more of plans **1012** to generated refined OBO **1018**. Refined OBO **1018** may be provided to RCCP service **1010** for refined OBO **1018** to be solved to produce plans **1012** once more. This loop can be iteratively repeated until goal inspector **1016** determines the business goals have been sufficiently achieved. In embodiments, additional Goals **1** through **n** **1014** may be provided to goal inspector **1016** to add on to the requirements for solving refined OBO **1018**.

(83) Once goal inspector **1016** determines plans **1012** have sufficiently achieved the goals as defined in OBO asks **1008** or goals **1** through **n** **1014**, goal inspector **1016** can pass solved plans **1020** to upsert service **1022**. Upsert service **1022** may then sent plans **1020** to database **1024** which may also be called a persistence layer. Plans **1020** may thereafter be used to perform various analytics at **1026**, at which point map-reduce type approach **1000** ends.

(84) Embodiments provide for an iterative technique to inspect the business objectives, and keep iterating until the results are closer to the goals. In addition, or as an alternative, once embodiments got closer to the goals, then the simulations are stopped, and it is sent to a persistence layer to build meaningful analytics.

(85) FIG. **11** illustrates user interface **1100**, according to an embodiment. User interface **1100** may be generated by user interface module **218** of supply chain planner **110**. User interface **1100** comprises various panes and windows indicating data related to scenarios generated by supply chain planner **110**. In this example, a top pane of user interface **1100** defines a risk within the supply chain planning solution of “Shift in consumer buying behavior to low grammage products,” with a confidence level of 86%.

(86) A lower pane of user interface **1100** lays out various scenarios that may be selected by a user interacting with user interface **1100**. As shown, the scenarios of the lower pane have been optimized by “Margin MAX” and “Demand Met MAX.” Scenario **1** **1102** is shown as a recommended scenario of 3 generated scenarios compared to a current plan. Each of the three scenarios, as well as the current plan, show a projected revenue, a projected gross profit margin, and a projected service level. As shown, scenario **1** **1102** projects an improvement, compared to the current plan, in revenue and gross profit margin, while maintaining the projected cost to serve and the projected service level. User interface **1100** includes an “Accept” button underneath scenario **1**, allowing a user to select scenario **1** for implementation.

(87) FIG. **12** illustrates user interface **1200**, according to an embodiment. User interface **1200** shows a comparison of scenario **3** **1204** and scenario **1** **1206** for various time periods **1202**. Scenarios **1204** and **1206** have been compared according to “Projected MAT revenue.” As shown by the buttons at the top of user interface **1200**, a user may select to compare scenario **1**, scenario **2** or scenario **3** against the current plan, while including or excluding any individual scenario. The user also may select to compare the scenarios to the current plan according to other metrics via the drop down menu, such as revenue, gross profit margin, cost to serve, service level, or any other relevant metric.

(88) FIG. **13** illustrates example comparison chart **1300**, according to an embodiment. Chart **1300** shows a comparison of revenue (y-axis **1302**) for various time periods (x-axis **1304**) for current plan **1306** and recommended scenario **1308**. In addition, or as an alternative, embodiments provide for several other ways of comparing the results, as shown in the above figure. As these scenarios are run, this output is displayed on a graphical user interface.

(89) Embodiments provide for an exemplary use case for a potato chips company. As an example

only and not by way of limitation, consider the following example to demonstrate the operation of the systems and methods described above. Assume there is a food plant that is producing potato chips and bagging them into a first bag size, and a second, smaller, bag size, before they are shipped into a distribution network. In this example, one challenge is that consumer buying behavior has shifted to favor smaller size bag keeping units (SKUs), although the larger bag SKUs are being stocked. Continuing with this example, the demand for the larger sized bags has decreased but are still being produced, because of the way the supply chain is configured. That is, once the potato chips come out of the fryer, they have to be bagged into either the large or smaller sized SKUs. In this example, although there is reduced demand for larger size bag, they are still being produced and shipped into the distribution network at the same rate. Once in the network, the larger size bags may be placed in inventory without significant sales, leading to eventual waste of the products. On the other hand, the smaller bag SKUs, where the demand has increased due to the change in consumer buying behavior, the food plant is barely able to keep up with the demand, and customer service becomes a problem.

(90) In addition, or as an alternative, there is a related contract risk in this example, where all the manufacturing is happening at a contract manufacturer. That is, the contract manufacturer has been promised with a certain volume across these larger and smaller size bag SKUs, so a simple solution of producing less of the larger size bag would likely violate the contract.

(91) In terms of input variables, embodiments allow perturbations of levers in this kind of use case, which will have a positive impact on the demand. Likewise offering some discounts or other promotions may have a result of increasing demand for the larger size bag, which will allow for the depletion of some of the inventory that is being built up. Other levers could be, for example, to make less potato chips, so then there will be less to bag and then less to ship. Another lever to look at may be to change some of the larger size bag lines to smaller size bag lines. According to embodiments, these may be the input variables where there does not need to specify what those exact levers are, or those exact impacts are. Rather, a user may specify the levers and embodiments automatically respond with what the best outcome is.

(92) As an example only and not by way of limitation, consider the following further examples to demonstrate the use of the systems and methods described above. A demand planner may use embodiments in order to fulfill his primary goal, which may be, for example, to develop driver-based monthly volume forecasts for products within supply chain network **100**. When performing demand planning related tasks, embodiments may use supply chain planner **110** to generate driver-based forecasts and identify demand shifts and related obsolescence risk resulting from such shifts for certain SKUs, (such as SKUs above a certain weight threshold.) A demand planner can raise a potential risk manually and request for input from the sales and marketing team to mitigate the risk. In such a situation, embodiments provide end to end workflow tracking for timely decision making.

(93) A marketing lead, a sales lead or a category lead may use embodiments in order to fulfill their primary goals, which may be, for example, to review and refine market demand assumptions within supply chain network **100**. These roles may participate in a variety of activities which are all grouped under the umbrella of demand review. For example, during demand review, a marketing lead may review category performance against a target and identify changes in demand shift, a sales lead may conduct pricing and promotion analysis with a, for example, net revenue management (NRM) team to assess the need to add or remove any promotions, pricing, etc. specific to these packs, and a category lead for a product analyzes the impact of demand shift in the plan and against the overall financial targets/metrics (e.g., revenue, margin) and prioritizes the mitigation of revenue risks. Also during demand review, a demand planner may create requisite simulations using embodiments, using pricing or promotion levers suggested by embodiments, to increase the demand for certain products along with their associated financial impacts. Finally, during demand review, a supply chain director and supply planner may be invited to help with the mitigation options. Embodiments may be used during demand review to provide end to end

workflow tracking for timely decision making.

(94) A supply chain director may use embodiments in order to fulfill his primary goal, for example, to evaluate supply options along with financial and business impact and decide trade offs. A supply chain director, during supply review, may use embedded analytics to articulate the product mix impact and call out additional impacts, such as a shift in bottlenecks within supply chain network **100** or risks of not meeting contractual obligations with third party manufacturers. The supply chain director may also suggest strategies around controlling technology throughout to balance between excess of certain products and shift towards alternative products. A supply chain planner may use guidance from the supply chain director and use embodiments to prepare requisite simulations with cost benefit analysis, projected customer service level basis, planned utilization ranges and inventory (min or max) levels. The supply chain director may pick a preferred option from the output of the simulations and decide to bubble up the decision for visibility to executives

(95) Supply chain executives may use embodiments in order to fulfill their primary goals, for example, to review and sign off on updated business plans. A supply chain executive, during management business review, may review the decisions taken by lower level employees and approve the business plan, and may also communicate the plan to stakeholders. The supply chain executive may use embodiments to provide end to end workflow tracking for timely decision making, as well as to allow management to review decisions taken in the next cycle.

(96) Reference in the foregoing specification to “one embodiment”, “an embodiment”, or “some embodiments” means that a particular factor, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

(97) While the exemplary embodiments have been shown and described, it will be understood that various changes and modifications to the foregoing embodiments may become apparent to those skilled in the art without departing from the spirit and scope of the present invention.

Claims

1. A system for performing rough cut capacity planning through iterative simulation, comprising: a supply chain network comprising two or more supply chain entities modelled as a set of nodes and edges; a computer coupled with a database and comprising a processor and memory, the computer configured to: receive supply chain transaction data as transaction tables; generate a base plan; generate updated transaction tables; denormalize the base plan and the updated transaction tables; generate supply chain network flow paths and supply chain network data; solve a rough cut capacity planning problem based at least in part on the supply chain network flow paths, the supply chain network data and simulation data; repeat at least the generate supply chain network flow paths and supply chain network data and the solve the rough cut capacity planning problem until business goals of the rough cut capacity planning meet a threshold; and update the simulation data based on an upsert process; and automated robotic production machinery configured to: produce one or more products based, at least in part, on the solution to the rough cut capacity planning problem.

2. The system of claim 1, wherein the computer is further configured to: denormalize the updated transaction tables by adding redundant data to transaction data to improve a read performance of the transaction data.

3. The system of claim 1, wherein the upsert process further comprises: update a corresponding entry in the database; and in response to the corresponding entry not existing, insert a new entry into the database.

4. The system of claim 1, wherein the supply chain network flow paths and the supply chain network data are generated at least in part based on a dataframe, the dataframe comprising a set of

the nodes and edges of the supply chain network, the denormalized base plan and the denormalized updated transaction tables.

5. The system of claim 1, wherein the computer is further configured to: relax one or more supply chain network constraints; invert the supply chain network of nodes and edges; and traverse a perturbation in the one or more supply chain network constraints as one or more demands in a reverse direction towards one or more customer nodes.

6. The system of claim 1, wherein the computer is further configured to: determine one or more estimates of production in order to meet one or more simulated demands.

7. The system of claim 1, wherein the rough cut capacity planning comprises one or more planning scenarios.

8. A computer-implemented method for performing rough cut capacity planning through iterative simulation, comprising: receiving, by a computer comprising a processor and memory and coupled to a database, supply chain transaction data as transaction tables; generating, by the computer, a base plan; generating, by the computer, updated transaction tables; denormalizing, by the computer, the base plan and the updated transaction tables; generating, by the computer, supply chain network flow paths and supply chain network data; solving, by the computer, a rough cut capacity planning problem based at least in part on the supply chain network flow paths, the supply chain network data and simulation data; repeating, by the computer, at least the generating supply chain network flow paths and supply chain network data and the solving the rough cut capacity planning problem until business goals of the rough cut capacity planning meet a threshold; updating, by the computer, the simulation data based on an upsert process; and producing, by automated robotic production machinery, one or more products based, at least in part, on the solution to the rough cut capacity planning problem.

9. The method of claim 8, further comprising: denormalizing, by the computer, the updated transaction tables by adding redundant data to transaction data to improve a read performance of the transaction data.

10. The method of claim 8, wherein the upsert process further comprises: updating, by the computer, a corresponding entry in the database; and in response to the corresponding entry not existing, inserting, by the computer, a new entry into the database.

11. The method of claim 8, wherein the supply chain network flow paths and the supply chain network data are generated at least in part based on a dataframe, the dataframe comprising a set of the nodes and edges of the supply chain network, the denormalized base plan and the denormalized updated transaction tables.

12. The method of claim 8, wherein performing the rough cut capacity planning (RCCP) further comprises: relaxing, by the computer, one or more supply chain network constraints; inverting, by the computer, the supply chain network of nodes and edges; and traversing, by the computer, a perturbation in the one or more supply chain network constraints as one or more demands in a reverse direction towards one or more customer nodes.

13. The method of claim 8, wherein performing the rough cut capacity planning (RCCP) further comprises: determining, by the computer, one or more estimates of production in order to meet one or more simulated demands.

14. The method of claim 8, wherein the rough cut capacity planning comprises one or more planning scenarios.

15. A non-transitory computer-readable storage medium embodied with software for performing rough cut capacity planning through iterative simulation, the software when executed by a computer comprising a processor and memory and coupled to a database, the software configured to: receive supply chain transaction data as transaction tables; generate a base plan; generate updated transaction tables; denormalize the base plan and the updated transaction tables; generate supply chain network flow paths and supply chain network data; solve a rough cut capacity planning problem based at least in part on the supply chain network flow paths, the supply chain

network data and simulation data; repeat at least the generate supply chain network flow paths and supply chain network data and the solve the rough cut capacity planning problem until business goals of the rough cut capacity planning meet a threshold; update the simulation data based on an upsert process; and produce, by automated robotic production machinery, one or more products based, at least in part, on the solution to the rough cut capacity planning problem.

16. The non-transitory computer-readable storage medium of claim 15, wherein the software when executed is further configured to: denormalize the updated transaction tables by adding redundant data to transaction data to improve a read performance of the transaction data.

17. The non-transitory computer-readable storage medium of claim 15, wherein the upsert process further comprises: update a corresponding entry in the database; and in response to the corresponding entry not existing, insert a new entry into the database.

18. The non-transitory computer-readable storage medium of claim 15, wherein the supply chain network flow paths and the supply chain network data are generated at least in part based on a dataframe, the dataframe comprising a set of the nodes and edges of the supply chain network, the denormalized base plan and the denormalized updated transaction tables.

19. The non-transitory computer-readable storage medium of claim 15, wherein the software when executed is further configured to: relax one or more supply chain network constraints; invert the supply chain network of nodes and edges; and traverse a perturbation in the one or more supply chain network constraints as one or more demands in a reverse direction towards one or more customer nodes.

20. The non-transitory computer-readable storage medium of claim 15, wherein the software when executed is further configured to determine one or more estimates of production in order to meet one or more simulated demands.
